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CHEMICAL SHIFTS IN THE NUCLEAR MAGNETIC RESONANCE SPECTRA OF MOLECULES CONTAINING POLAR GROUPS[1]

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ABSTRACT

It is shown that the electric field at a particular nucleus arising from polar groups in other parts of a molecule can lead to chemical shifts proportional to the first power of the field strength. The change $\Delta\sigma$ in the proton screening constant of an X—H bond when it is subjected to an electric field E is approximately given by

$$\Delta\sigma = -2\times10^{-12}E_z - 10^{-18}E^2$$

where E_z is the component of E in the bond direction. A field along the X—H bond draws the electrons in the enriched region between the nuclei away from the proton, thereby causing its resonance to occur at lower magnetic field strengths, while a field in the H—X direction leads to resonance at higher fields. The electric field produced by the polarization of neighboring solvent molecules may also be important and lead to a solvent shift related to the dielectric constant of the solvent. The direction of the internal field, and that due to the solvent, can easily be estimated when the molecule is rigid, so that the model can be useful in assigning spectra. It gives a simple and reasonably accurate account of the observed spectra of substituted benzenes and of some solvent effects on the proton resonances of *cis*- and *trans*-1,2-dichloroethene.

INTRODUCTION

In nuclear magnetic resonance experiments, nuclei with magnetic moments are effectively used as probes to measure the actual magnetic field, H_{local}, at the nucleus.

$$H_{\text{local}} = (1-\sigma)H \tag{1}$$

where H is the applied magnetic field strength and σ is the *screening constant*. Owing to rapid molecular motion in gases and liquids, the observed screening constant corresponds to an average for all molecular orientations. The molecules which are far removed from the one with the resonant nucleus interact with it as if they formed a continuum having the bulk properties of the fluid, and they may contribute to σ a term proportional to the volume susceptibility; this term depends on the sample shape, and vanishes for a spherical specimen. The screening constants quoted in this paper have all been "corrected" so as to remove this bulk susceptibility effect.

Progress in understanding observed screening constants has largely been based on models in which σ is split up into atomic contributions (see Pople, Schneider, and Bernstein (1) and Saika and Slichter (2)). Thus,

$$\sigma = \sigma_d + \sigma_p + \sigma_o + \sigma_c \tag{2}$$

where σ_d and σ_p are the dia- and para-magnetic shielding of the nucleus whose total screening constant is σ; σ_d is proportional to the magnetic field produced at the nucleus by the moment induced through the rotation of the electrons about the nucleus with the Larmor precessional frequency $eH/2mc$. The paramagnetic shielding σ_p arises from the hindrance to the Larmor precession produced by the lack of axial symmetry about the field H; in a linear molecule, σ_p is zero when H is along the molecular axis, but in general it is non-zero for an atom in a molecule. σ_o is the shielding constant proportional to the field produced by the magnetic moments induced in other atoms, and σ_c is the contribution

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from interatomic currents (3); σ_0 is zero in a tumbling molecule if the other atoms have isotropic susceptibilities.

A uniform electric field E_z distorts an atom in an S-state, and this distortion affects σ_d and σ_p. σ_d is diminished in both the z and x (or y) directions, the diminution being greater in the z-direction; σ_p is zero in the z-direction, but is negative in the x- and y-directions. The change in σ due to E_z is proportional to E_z^2, for the symmetry of the atom requires the shielding to be unaffected by a reversal of E_z. For a hydrogen atom in its ground state, Marshall and Pople (4) showed that, neglecting the effect of electron spin,

$$\sigma_{zz} = \frac{e^2}{3mc^2a}\left\{1-\frac{439}{40}\frac{a^4E_z^2}{e^2}\right\}$$

[3]

$$\sigma_{xx} = \sigma_{yy} = \frac{e^2}{3mc^2a}\left\{1-\frac{193}{15}\frac{a^4E_z^2}{e^2}\right\}$$

for weak electric fields ($E_z \ll e/a^2$), where e is the protonic charge (4.803×10^{-10} e.s.u.) and a the Bohr radius (0.5292 Å).

If the resonant nucleus is not at a molecular center of inversion, the screening for a fixed orientation of the molecule may be affected by a reversal of E_z, so that in weak fields σ may be proportional to E_z, rather than E_z^2. If E_z is a fixed external field, the shielding proportional to E_z averages to zero in a gas or liquid, but if E_z arises from a polar group within the molecule itself, or from neighboring solvent molecules polarized by the solute, then the mean value of E_z at a particular nucleus may be non-zero. A screening constant proportional to E_z, in addition to one proportional to E_z^2, is therefore possible. Thus in an X—H bond, σ for the proton will be proportional to the first power of the component of the field along the bond, and to the second power of the field perpendicular to it.

The purpose of this paper is to investigate the influence of this "internal field" effect on proton screening constants in gaseous molecules, and to discuss changes in σ arising from fields due to the presence of a solvent; the latter part extends Stephen's approach (5) to the problem of solvent effects.

GENERAL THEORY

In the presence of a primary uniform magnetic field $\mathbf{H}$, there will be, at a particular nucleus, an induced secondary magnetic field $\mathbf{H}'$, and the screening constant tensor $\sigma_{\alpha\beta}$ for this nucleus is defined by

[4] $$H'_\alpha = -\sigma_{\alpha\beta}H_\beta$$

where the Greek subscripts denote tensor components (α and β can be x, y, or z, and repeated subscripts indicate a summation over x, y, and z components; thus $\sigma_{\alpha\beta}H_\beta = \sigma_{\alpha x}H_x+\sigma_{\alpha y}H_y+\sigma_{\alpha z}H_z$). It is supposed that if the molecule is in a uniform electric field $\mathbf{E}$, the screening is affected in such a way that σ can be expanded as a power series in E

[5] $$\sigma_{\alpha\beta} = \sigma_{\alpha\beta}^{(0)}+\sigma_{\alpha\beta\gamma}^{(1)}E_\gamma+\tfrac{1}{2}\sigma_{\alpha\beta\gamma\delta}^{(2)}E_\gamma E_\delta+\ldots$$

If the molecule is symmetric in x (that is, if $\sigma_{\alpha\beta}$ is unaffected by an inversion in the yz plane), then $\sigma_{\alpha\beta x}^{(1)} = 0$. Thus $\sigma^{(1)} = 0$ for an atom; for a molecule which is symmetrical about the z-axis (an X—H bond is usually approximately symmetric about the bond direction) $\sigma_{\alpha\beta x}^{(1)} = \sigma_{\alpha\beta y}^{(1)} = 0$. The observed screening constant σ for a molecule in a gas or liquid corresponds to an average of $\sigma_{\alpha\beta}$ over all orientations. Averaging over all molecular

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orientations is equivalent to fixing the molecule (and in this case the electric field) and averaging over all directions of the magnetic field. Thus,

$$\sigma = \frac{1}{H^2}\sigma_{\alpha\beta}\overline{H_\alpha H_\beta} = \tfrac{1}{3}\sigma_{\alpha\alpha} = \tfrac{1}{3}\{\sigma^{(0)}_{\alpha\alpha}+\sigma^{(1)}_{\alpha\alpha\gamma}E + \tfrac{1}{2}\sigma^{(2)}_{\alpha\alpha\gamma\delta}E_\gamma E_\delta + \ldots\}. \tag{6}$$

For an atom in an S-state $\sigma^{(1)} = 0$ and there are two independent $\sigma^{(2)}$ tensor components. For a hydrogen atom in a uniform field E_z, [5] yields

$$\sigma_{zz} = \sigma^{(0)}+\tfrac{1}{2}\sigma^{(2)}_{zzzz}E_z^{\,2}+\ldots$$
$$\sigma_{xx} = \sigma^{(0)}+\tfrac{1}{2}\sigma^{(2)}_{xxzz}E_z^{\,2}+\ldots$$

whence, from [3],

$$\sigma^{(0)} = \frac{e^2}{3mc^2a}\,; \qquad \sigma^{(2)}_{zzzz} = \sigma^{(2)}_{||} = -\frac{439}{60}\frac{a^3}{mc^2} \tag{7}$$
$$\sigma^{(2)}_{xxzz} = \sigma^{(2)}_{\perp} = -\frac{386}{45}\frac{a^3}{mc^2}\,.$$

The $\sigma^{(2)}$ tensor is given by

$$\sigma^{(2)}_{\alpha\beta\gamma\delta} = \sigma^{(2)}_{\perp}\delta_{\alpha\beta}\delta_{\gamma\delta}+\tfrac{1}{2}(\sigma^{(2)}_{||}-\sigma^{(2)}_{\perp})(\delta_{\alpha\gamma}\delta_{\beta\delta}+\delta_{\alpha\delta}\delta_{\beta\gamma}) \tag{8}$$

where $\delta_{\alpha\beta}$ is the substitution tensor ($= 1$ if $\alpha = \beta$, $= 0$ if $\alpha \neq \beta$).

In an axially symmetric X—H bond there can be no screening proportional to the first power of the field at right angles to the bond. Thus $\sigma^{(1)}_{\alpha\beta x} = \sigma^{(1)}_{\alpha\beta y} = 0$, and there are two independent $\sigma^{(1)}$ tensors, $\sigma^{(1)}_{zzz} = \sigma^{(1)}_{||}$ and $\sigma^{(1)}_{xxz} = \sigma^{(1)}_{yyz} = \sigma^{(1)}_{\perp}$. From [6] and [8]

$$\sigma = \sigma^{(0)}+\sigma^{(1)}E_z+\tfrac{1}{2}\sigma^{(2)}E^2+\ldots \tag{9}$$

where

$$\sigma^{(0)} = \tfrac{1}{3}(\sigma^{(0)}_{||}+2\sigma^{(0)}_{\perp}) \qquad \text{and} \qquad \sigma^{(1)} = \tfrac{1}{3}(\sigma^{(1)}_{||}+2\sigma^{(1)}_{\perp}).$$

To estimate the importance of the proton shielding proportional to E and to E^2 in an X—H bond, use can be made of the results of Marshall and Pople (4) for the E^2 effect on an H-atom. Consider a model in which an H-atom is at a distance R from a point charge λ. In the presence of a uniform electric field **E**, the total electric field **F** acting on the atom is

$$\mathbf{F} = \{E_x, E_y, E_z+(\lambda/R^2)\} \tag{10}$$

and hence an effect proportional to F^2 will involve terms of the first and second order in E. The screening constant for the proton in this "molecule" is given by [5] and [8] and is

$$\sigma_{\alpha\beta} = \frac{e^2}{3mc^2a}\delta_{\alpha\beta}-\frac{193}{45}\frac{a^3F^2}{mc^2}\delta_{\alpha\beta}+\frac{227}{360}\frac{a^3}{mc^2}F_\alpha F_\beta. \tag{11}$$

Because of the form of F (see [10]), [11] leads to screening proportional to E_z and to E^2, and it can be shown that

$$\sigma^{(1)}_{||} = \frac{\lambda}{R^2}\sigma^{(2)}_{||} \qquad \text{and} \qquad \sigma^{(1)}_{\perp} = \frac{\lambda}{R^2}\sigma^{(2)}_{\perp}\,.$$

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Thus [9] becomes

$$\sigma = \frac{e^2}{3mc^2a} - \frac{881}{108}\frac{a^3}{mc^2}\frac{\lambda}{R^2}E_z - \frac{881}{216}\frac{a^3}{mc^2}E^2 - \ldots$$

[12] $$= \sigma^{(0)} + \sigma^{(2)}\left\{\frac{\lambda}{R^2}E_z + \tfrac{1}{2}E^2\right\} + \ldots$$

where

$$\sigma^{(2)} = \tfrac{1}{3}(\sigma^{(2)}_{||} + 2\sigma^{(2)}_{\perp}).$$

From [7], $\sigma^{(0)} = 17.8\times10^{-6}$, $\sigma^{(2)} = -1.5\times10^{-18}$ e.s.u., $\lambda = 10^{-10}$ e.s.u., and typical parameters are $R = 10^{-8}$ cm, whence

[13] $$\sigma = 2\times10^{-5} - 2\times10^{-12}E_z - 10^{-18}E^2 - \ldots$$

Thus for $E = 10^5$ e.s.u. (the field 7 Å from a proton), the shielding proportional to E is $\sim$0.2 p.p.m., and is $\sim$20 times that proportional to E^2.

The sign of $\sigma^{(1)}$ is opposite to that of λ. However, in most X—H bonds, it is reasonable to assume that there is an increased electronic charge between the nuclei (as for a positive λ), so that an electric field in the X—H direction will tend to draw this excess charge further away from the proton, thereby decreasing its shielding constant. Equation [13] indicates that the change in σ arising from the action of an electric field on an XH bond is approximately $-2\times10^{-12}E_z - 10^{-18}E^2$, and the measurements discussed later in this paper confirm this result in the case of CH bonds.

SOLVENT EFFECTS

When a polar molecule is dissolved, it polarizes the surrounding medium, and this polarization leads to an electric field—the "reaction field"—at the solute. If the molecule is sufficiently symmetrical, the mean reaction field is parallel and proportional to its dipole moment. For the Onsager model (6), in which the solute molecule is represented by a sphere of radius r containing a point dipole of moment μ at its center, and the solvent by a continuum of dielectric constant ϵ, the reaction field $\mathbf{R}$ is

[14] $$\mathbf{R} = \frac{2(\epsilon-1)}{2\epsilon+1}\frac{\mathbf{m}}{r^3} = \frac{2(\epsilon-1)(n^2-1)}{3(2\epsilon+n^2)}\frac{\boldsymbol{\mu}}{\alpha}$$

where $\mathbf{m} = \boldsymbol{\mu} + \alpha\mathbf{R}$ is the total dipole moment of the solute in the medium, $\alpha = \frac{n^2-1}{n^2+2}r^3$ being the polarizability of the sphere, and n the refractive index of the pure solute (as a liquid or solid). Thus $n^2 \sim 2.5$, and

[15] $$\mathbf{R} = \frac{\epsilon-1}{2\epsilon+2.5}\frac{\boldsymbol{\mu}}{\alpha}.$$

Protons are normally on the "surface" of a molecule and are therefore exposed to direct contact with the surrounding molecules, so that the Onsager field [14] is probably frequently a poor approximation to the actual reaction field acting on a proton, although it might adequately describe the field at a nucleus in the interior of a molecule. (Thus the C^{13} screening constants in substituted methanes $C^{13}X_nY_{4-n}$ should be approximately linearly dependent on $(\epsilon-1)/(2\epsilon+2.5)$.) Local solvent effects due to polar groups near a proton are probably important; thus in the non-dipolar molecule 1,4-dinitrobenzene, fields arising from the induced charges on the solvent molecules near the highly polar

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nitro groups probably lead to electric fields at the protons, even though $\mu = 0$. The reaction field gradient R' for a model similar to Onsager's is (7)

$$R' = \frac{6(\epsilon-1)}{3\epsilon+2}\frac{\Theta}{r^5} \qquad [16]$$

where Θ is the molecular quadrupole moment; for a molecule with two opposed dipolar groups of moment μ separated by a distance d, $\Theta = 2\,\mu d$. Figure 1 shows the probable

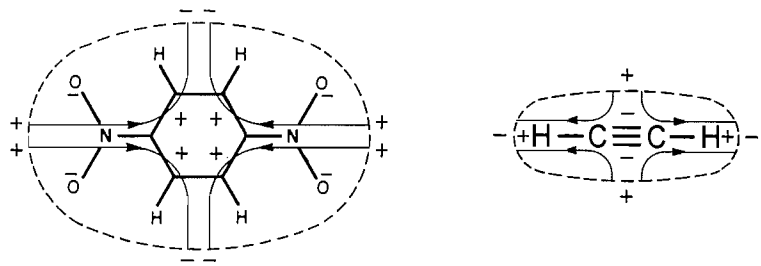

FIG. 1. Quadrupolar reaction fields in 1,4-dinitrobenzene and in acetylene.

form of the reaction field in 1,4-dinitrobenzene and in acetylene. If one of the protons in acetylene were replaced by an electronegative group (e.g., F, Cl, CN) a larger solvent shift would be expected, for in this case the shifts due to R and to R' are both to lower magnetic field strengths. When both R and R' are important, [15] and [16] indicate that σ should be approximately linearly dependent on $(\epsilon-1)/(\epsilon+1)$.

DISCUSSION

Although the well-known inductive and mesomeric effects undoubtedly play an important part in determining charge distributions in substituted aromatic molecules, the protons are isolated from the ring by sigma bonds, making it dangerous to try to correlate proton signals with pi-electron densities on the associated carbon atoms, or with Hammett acidity functions. In fact, it is shown in this section that the general pattern of proton screening constants in mono- and poly-substituted benzenes can be largely explained in terms of [13] and "internal" and "reaction" fields.

Consider a substituted benzene C_6H_5X, in which there is a "bond-dipole" μ as shown. If μ is situated at a point distant half-way along the X—C bond, and if the benzene ring is a regular hexagon, then the electric fields at the five protons, in the directions of the various CH bonds, are (if μ is the only significant contributor to the field)

$$E_{CH_2} = E_{CH_6} = \frac{4\mu\{10l_{CC}^2-8l_{CH}^2-2l_{CX}^2+10l_{CC}l_{CH}+5l_{CC}l_{CX}+13l_{CH}l_{CX}\}}{\{4l_{CC}^2+4l_{CH}^2+l_{CX}^2+4l_{CC}l_{CH}+2l_{CC}l_{CX}-2l_{CH}l_{CX}\}^{5/2}}$$

$$E_{CH_3} = E_{CH_5} = \frac{4\mu\{42l_{CC}^2+8l_{CH}^2+2l_{CX}^2+42l_{CC}l_{CH}+21l_{CC}l_{CX}+13l_{CH}l_{CX}\}}{\{12l_{CC}^2+4l_{CH}^2+l_{CX}^2+12l_{CC}l_{CH}+6l_{CC}l_{CX}+2l_{CH}l_{CX}\}^{5/2}} \qquad [17]$$

$$E_{CH_4} = \frac{16\mu}{\{4l_{CC}+2l_{CH}+l_{CX}\}^3}$$

where l_{AB} is the length of the A—B bond. For $l_{CC} = 1.39$ Å, $l_{CH} = 1.07$ Å, $l_{CX} = 2.0$ Å

$$\begin{aligned} E_{CH_2} &= 8.74\times10^{22}\,\mu; & E_2^2 &= 82.8\times10^{44}\,\mu^2, \\ E_{CH_3} &= 2.32\times10^{22}\,\mu; & E_3^2 &= 5.88\times10^{44}\,\mu^2, \\ E_{CH_4} &= 1.75\times10^{22}\,\mu; & E_4^2 &= 3.07\times10^{44}\,\mu^2. \end{aligned} \qquad [18]$$

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Thus for an isolated nitrobenzene molecule, $\mu \sim 4\times10^{-18}$ e.s.u., so that [13] leads to the following contributions to σ arising from E_z and E^2

$$\begin{aligned} \sigma_{H_2} &= \sigma_{H_6} = -0.70\times10^{-6}-0.13\times10^{-6}, \\ \sigma_{H_3} &= \sigma_{H_5} = -0.19\times10^{-6}-0.01\times10^{-6}, \\ \sigma_{H_4} &= -0.14\times10^{-6}-0.005\times10^{-6}. \end{aligned} \tag{19}$$

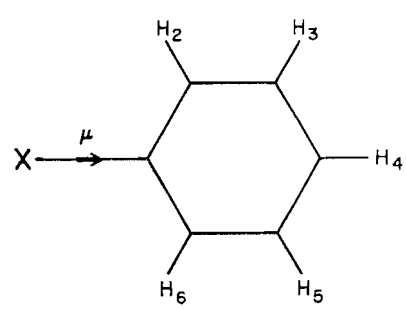
From [19], it is clear that the approximately isotropic E^2 effect is of secondary importance, and is negligible except where the "source" of the electric field is close to the proton (as, for example, in hydrogen bonds (8)). If the electric field effects are the only significant contributors to the relative chemical shifts, [19] indicates that in gaseous nitrobenzene, for example, the *o*-protons should have the largest shift to low field (relative to benzene itself), followed by the *m*- and *p*-hydrogens. Unfortunately, there are no data on gaseous substituted benzenes, but in solution in cyclohexane (9) the *o*-proton in $C_6H_5NO_2$ is shifted by -0.97, the *m*, -0.30, and the *p*, -0.42 p.p.m. relative to benzene in the same solvent.

The reaction field R of [15] tends to shift the H_3 and H_4 signals to low field (H_3 being influenced less than H_4) and H_2 to high field. These data (9) were obtained in a solvent which was a mixture of equal volumes of cyclohexane and nitrobenzene, for which $\epsilon = 16$, so that for $\mu/\alpha = 4.0\times10^5$, $R = 1.7\times10^5$ e.s.u. When E_z in [13] is put equal to $R\cos\phi$, where ϕ is the angle between **R** and a C—H axis, the following solvent shifts are obtained:

$$(\sigma_{H_2})_s = 0.14\times10^{-6}, \quad (\sigma_{H_3})_s = -0.20\times10^{-6}, \quad (\sigma_{H_4})_s = -0.37\times10^{-6}. \tag{20}$$

Combining [19] and [20], it is found that the *o*-, *m*-, and *p*-protons should be shifted by -0.69, -0.40, and -0.51 p.p.m. relative to benzene, in qualitative agreement with the above facts. If this is a valid interpretation of the $C_6H_5NO_2$ spectrum, polar aliphatic solvents (e.g., acetone, acetonitrile) should increase the separation between the *m*- and *p*-proton signals, and bring the *o*-peak closer to the *p*-one (relative to the situation in a non-polar solvent like CCl_4).

If the sign of μ is changed (as, for example, in $C_6H_5NH_2$) E is reversed and shifts to higher field are expected. Also, since the E and E^2 effects now act in opposite directions, the *o*-signal should be closer to the *m*- and *p*-ones. These expectations are in accord with the facts (10).



The proton resonance spectrum of di- and tri-substituted benzenes should be interpretable in terms of the model. Thus in 1,2-dichlorobenzene, the difference between the electric fields acting along the CH_3 and CH_4 axes is (from [18]) $6.99\times10^{22}\,\mu$, and for a C—Cl bond dipole moment of 1.7×10^{-18} e.s.u., [13] leads to a difference in the resonance

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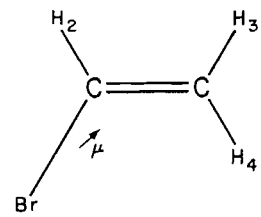
frequencies of the H_3 and H_4 protons of -0.24 p.p.m. The observed value is -0.25 p.p.m. for a solution in CCl_4 (11). The model predicts that increasing the polarity of the solvent will not affect H_3, but will shift the H_4 signal to lower field. Similarly, in 1,2,3-trichlorobenzene in CCl_4, the signal from H_4 and H_6 should be shifted -0.22 p.p.m. relative to that from H_5, but this difference should decrease as the polarity of the solvent increases. The observed shift in CCl_4 is -0.23 p.p.m. (11).

The effects of solvents on the chemical shifts of the protons in the 2- and 3-positions in some 1,4-disubstituted benzenes have been investigated by Schaefer and Schneider (12). They found that in *p*-methyl nitrobenzene the *m*-proton screening constant is shifted by -0.29 p.p.m. on changing the solvent from *n*-hexane to acetone. The corresponding shift for the *o*-proton is -0.06 p.p.m. Thus, although the *o*-proton is closer to the polar NO_2 group, its solvent shift is smaller. Our model is capable of explaining this, for the reaction field **R** is parallel to the 1-4 axis and therefore leads to a *m*-proton screening constant of $\frac{1}{2}\sigma^{(1)}R$ (that is, a shift to lower field, for $\sigma^{(1)}$ is negative) and an ortho shift of the same magnitude to higher field. (The small observed negative solvent shift (-0.06 p.p.m.) for the *o*-proton is partly due to an R^2 effect (this leads to a relative shift of -0.02 p.p.m.) but chiefly to the inadequacy of [15] for describing the reaction field in the immediate neighborhood of a highly polar group.) The difference between the *o*- and *m*-proton shifts in changing the solvent from *n*-hexane to acetone (-0.23×10^{-6}) should be approximately given by

$$\left\{\left(\frac{\epsilon-1}{2\epsilon+2.5}\right)_{\text{acetone}}-\left(\frac{\epsilon-1}{2\epsilon+2.5}\right)_{\text{hexane}}\right\}\frac{\mu}{\alpha}\sigma^{(1)}.$$

Taking $\mu/\alpha = 3.3\times10^5$ e.s.u., this leads to $\sigma^{(1)} = -2.6\times10^{-12}$ (confirming the order of magnitude of the estimate in [13]). However, our model cannot explain the large positive shift of the *m*-proton relative to the *o*-proton on changing the solvent from *n*-hexane to benzene. It is clear that some other type of interaction between *p*-methyl nitrobenzene and benzene must exist; Schaefer and Schneider (12) postulated hydrogen bonding between the *m*-proton and the π-electrons of the neighboring benzene molecule.

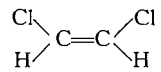
The model should sometimes be helpful in assigning observed peaks to particular protons. Thus in vinyl bromide, the direction of the "bond" dipole moment is probably approximately along the line joining the bromine nucleus to the mid-point of the C=C bond. The reaction field is therefore approximately parallel to the CH_3 and



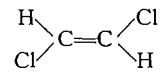
perpendicular to the CH_4 bonds. Hence, it can be anticipated that the H_3 resonance signal will be shifted more (to low field) than that of H_4 on increasing the dielectric constant of the aliphatic solvent.

In *cis*- and *trans*-1,2-dichloroethene (I and II) "local" effects are probably approximately the same for all protons, whereas only I has an effect due to a dipolar reaction field. Thus the protons in I should have a larger low-field shift on changing the solvent

from hexane to acetone; since μ/α for I is 2.3×10^5, the expected difference is -0.07 p.p.m.; the experimental value is -0.08 p.p.m. (13).



I



II

Equation [16] for the quadrupolar reaction field implies that appreciable solvent shifts will occur in non-dipolar molecules like 1,4-dinitrobenzene, for which $\Theta \sim -20\times10^{-26}$ e.s.u. The four protons will experience a reaction field $R = -(R'r \cos 30°)/2$ along the C—H bonds, and this should lead to a shift of -0.15 p.p.m. on changing the solvent from *n*-hexane to acetone. In NNN′N′-tetramethyl *p*-phenylene diamine Θ will be positive, so the corresponding shift should be positive. For acetylene, $\Theta \sim 3\times10^{-26}$ e.s.u. and $R = R'r$, so that the corresponding shift in the proton signals should be -0.19 p.p.m. Thus protons near polar groups experience "local" solvent electric fields which may be comparable to those due to the reaction fields of polar molecules. Fortunately both are approximately proportional to $(\epsilon-1)/(\epsilon+1)$, so that if other solvent influences (such as shape effects coupled with diamagnetic anisotropy, as in benzene or diacetylene, or strong van der Waals forces, as in CH_2I_2, or strong specific interactions, as in hydrogen bonded substances (14)) are absent σ should vary approximately linearly with this solvent parameter.

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